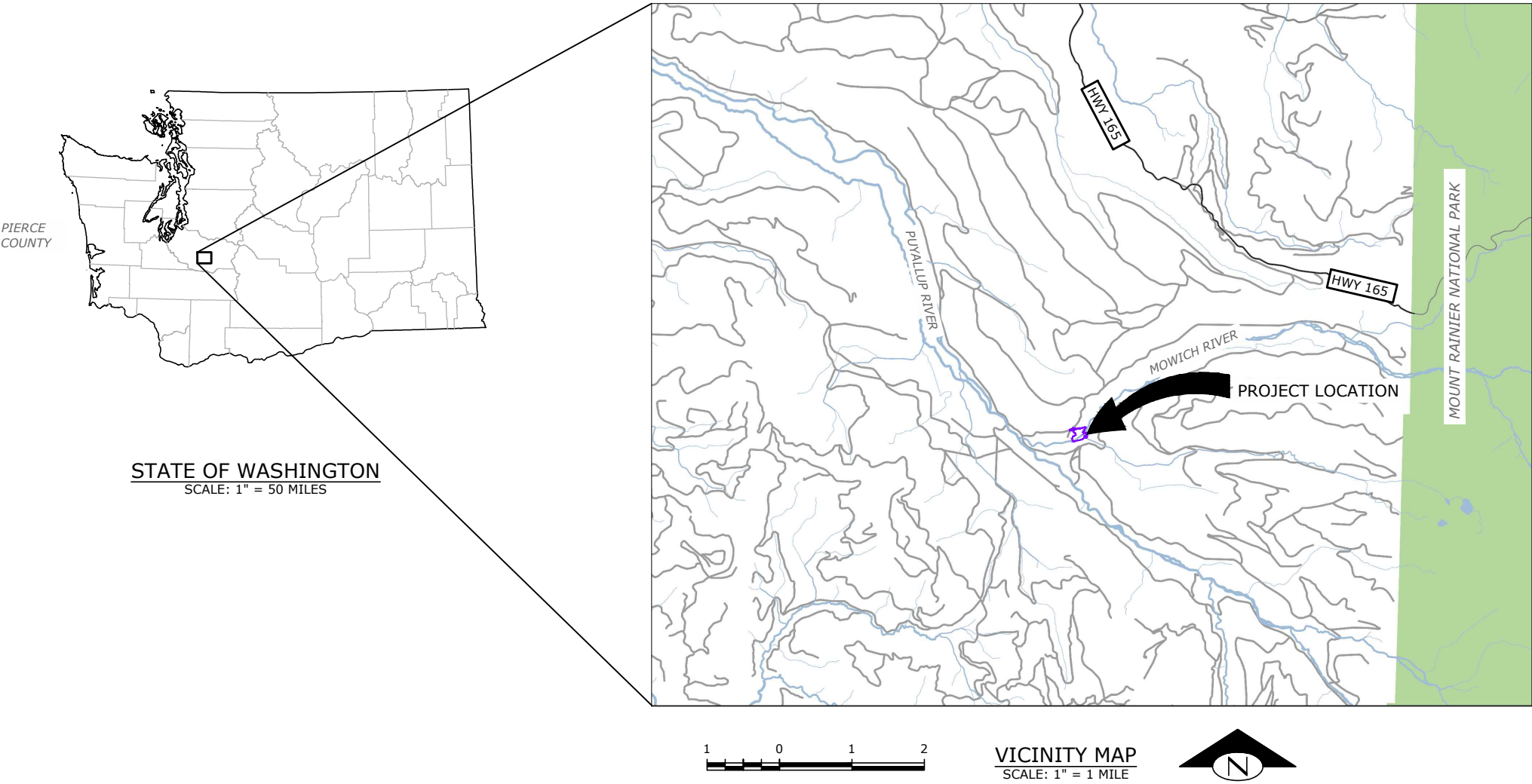


RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION

PRELIMINARY DESIGN

SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP



SHEET INDEX	
SHEET NUMBER	SHEET TITLE
1	COVER
2	GENERAL NOTES
3	LEGEND
4	EXISTING CONDITIONS PLAN
5	ACCESS AND STAGING OVERVIEW
6	ACCESS AND STAGING PLAN
7	ACCESS AND TESC DETAILS
8	PROPOSED CONDITIONS PLAN
9	CROSS SECTIONS
10	WOOD CATCH ELJ DETAIL
11	LEFT BANK ELJ DETAIL
12	ELJ DETAILS
13	RESTORATION PLAN

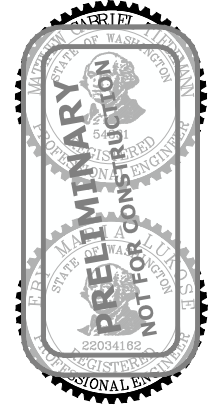
CONTACT INFORMATION

ENGINEER:

NATURAL SYSTEMS DESIGN, INC
1900 N NORTHLAKE WAY, SUITE 211
SEATTLE, WA 98103
(206) 834-0175

CONTRACTING AGENCY:

SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP
6700 MARTIN WAY EAST, SUITE 112
OLYMPIA, WA 98516



SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP
RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION
COVER
DRAFT PRELIMINARY DESIGN

DATE

APRIL 2026

COUNTY

PIERCE

LATITUDE

46°54'04"N

LONGITUDE

122°00'45"W

TN/SC/RG

T1.6N/S02/R06E

DESIGN MT. EL.

MP

CHECK

SW

CHECK

EL

01

IF THIS BAR DOES NOT MEASURE 1" THEN DRAWING IS NOT PLOTTED TO ORIGINAL SCALE.

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GENERAL NOTES

- 1. THESE PLANS HAVE BEEN PREPARED FOR THE EXCLUSIVE USE OF SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP, HEREAFTER REFERRED TO AS "OWNER", AND THEIR AUTHORIZED AGENTS.
- 2. NATURAL SYSTEMS DESIGN, HEREAFTER REFERRED TO AS "ENGINEER" IS RESPONSIBLE FOR THE PREPARATION OF THESE ORIGINAL PLANS AND ASSOCIATED SPECIFICATIONS; AND WILL NOT BE RESPONSIBLE FOR, OR LIABLE FOR, UNAUTHORIZED CHANGE, OR USE, OF THESE PLANS WHICH INCLUDES ALTERATION, DELETION, OR EDITING OF THIS DOCUMENT WITHOUT EXPLICIT WRITTEN PERMISSION FROM THE ENGINEER. ANY OTHER UNAUTHORIZED USE OF THIS DOCUMENT IS PROHIBITED.
- 3. THE LOCATION OF ALL FEATURES SHOWN IS APPROXIMATE. THE MOWICH RIVER IS A DYNAMIC SYSTEM AND HAS CHANGED SINCE THE DATE OF LIDAR COLLECTION. THE CONTRACTOR SHALL ANTICIPATE CHANGES IN THE RIVER PLANFORM BETWEEN BID AND CONSTRUCTION. FINAL LOCATIONS AND ELEVATIONS FOR ALL ELJS SHALL BE STAKED IN THE FIELD BY THE ENGINEER AND MAY CHANGE FROM THE LOCATIONS AND ELEVATIONS SHOWN IN THESE PLANS.

PERMIT NOTES

- 1. THE CONTRACTOR SHALL CONDUCT THE ACTIVITIES SHOWN IN THESE PLANS IN A MANNER THAT MINIMIZES THE ADVERSE IMPACT ON WATER QUALITY, FISH AND WILDLIFE, AND THE NATURAL ENVIRONMENT.
- 2. ALL WORK SHALL BE IN COMPLIANCE WITH PERMIT CONDITIONS. IT IS THE CONTRACTOR'S RESPONSIBILITY TO HAVE COPIES OF ALL PERMITS ON THE JOB SITE, UNDERSTAND AND COMPLY WITH ALL PERMIT CONDITIONS.
- 3. IF AT ANY TIME FISH ARE OBSERVED IN DISTRESS, A FISH KILL OCCURS, OR WATER QUALITY PROBLEMS DEVELOP (INCLUDING EQUIPMENT LEAKS OR SPILLS), OPERATIONS SHALL CEASE AND THE OWNER SHALL BE NOTIFIED IMMEDIATELY.
- 4. AVOID AND MINIMIZE ADVERSE IMPACTS TO WATERS OF THE UNITED STATES, INCLUDING MINIMIZING THE NUMBER, DURATION, AND EXTENT OF WORK BELOW ORDINARY HIGH WATER AND EQUIPMENT CROSSINGS OF WETTED CHANNELS.
- 5. IF, DURING CONSTRUCTION, ARCHAEOLOGICAL REMAINS ARE ENCOUNTERED, CONSTRUCTION IN THE VICINITY SHALL BE HALTED, AND THE STATE OFFICE OF HISTORIC PRESERVATION AND THE OWNER SHALL BE NOTIFIED IMMEDIATELY.

SURVEY NOTES

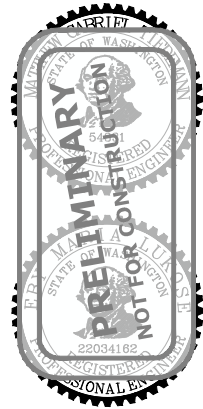
- 1. LIDAR FOR THIS PROJECT WAS COLLECTED IN 2022 AND IS REPRESENTATIVE OF 2022 CONDITIONS. THE VERTICAL DATUM IS NAVD88 (FT). THE HORIZONTAL DATUM IS NAD83 WASHINGTON STATE PLANE SOUTH AND THE UNIT IS US SURVEY FEET.
- 2. GATES, FENCELINES, AND UTILITIES WERE NOT SURVEYED. CONTRACTOR TO VERIFY IN FIELD.
- 3. PARCEL BOUNDARIES ARE FROM PIERCE COUNTY GIS AND ARE NOT SURVEYED.
- 4. AERIAL IMAGERY WAS COLLECTED IN AUGUST 2025 (SPSSEG).

CONSTRUCTION NOTES

- 1. THE CONTRACTOR SHALL ASSUME SOLE AND COMPLETE RESPONSIBILITY FOR JOB SITE CONDITIONS DURING THE COURSE OF CONSTRUCTION OF THIS PROJECT, INCLUDING SAFETY OF ALL PERSONS AND PROPERTY.
- 2. CONSTRUCTION HOURS SHALL BE WEEKDAYS BETWEEN 7:00 A.M. AND 6:30 P.M. UNLESS PRIOR APPROVAL IS RECEIVED FROM THE OWNER.
- 3. ANY DISCREPANCIES ARE TO BE BROUGHT TO THE ATTENTION OF THE OWNER PRIOR TO PROCEEDING WITH THE WORK.
- 4. THE CONTRACTOR SHALL PROTECT ALL EXISTING STRUCTURES, VEGETATION, AND IMPROVEMENTS NOT INDICATED FOR REMOVAL.
- 5. THE CONTRACTOR SHALL KEEP THE JOB SITE CLEAN AND HAZARD FREE.
- 6. THE CONTRACTOR SHALL DISPOSE OF ALL DIRT, DEBRIS, AND RUBBISH GENERATED BY THE WORK. UPON COMPLETION OF WORK, CONTRACTOR SHALL REMOVE ALL MATERIAL AND EQUIPMENT NOT SPECIFIED AS REMAINING ON THE PROPERTY.
- 7. NO TREES OR VEGETATION SHALL BE REMOVED UNLESS NOTED ON THE PLANS OR SPECIFIED ON-SITE BY THE OWNER OR THE ENGINEER. NO GRADING SHALL TAKE PLACE WITHIN THE DRIP LINE OF TREES NOT TO BE REMOVED UNLESS OTHERWISE APPROVED.
- 8. THE CONTRACTOR SHALL MAINTAIN A SET OF PLANS ON THE JOB SHOWING "AS-CONSTRUCTED" CHANGES MADE TO DATE. UPON COMPLETION OF THE PROJECT, THE CONTRACTOR SHALL SUPPLY TO OWNER A SET OF PLANS, MARKED UP TO THE SATISFACTION OF THE OWNER, REFLECTING THE AS-CONSTRUCTED MODIFICATIONS.

ELJ NOTES

- 1. LOGS AND POSTS SHALL BE DOUGLAS FIR. RACKING MATERIAL SHALL BE DOUGLAS FIR OR WESTERN HEMLOCK.
- 2. LOGS SHALL HAVE SPECIFIED DIAMETERS AS MEASURED AT DBH, DEFINED AS 4.5 FEET ABOVE GROUND WHEN TREE WAS STANDING.
- 3. EXISTING WOODY MATERIAL AT THE STRUCTURE LOCATION SHALL BE MOVED OR PROTECTED FROM CONSTRUCTION ACTIVITIES AND THEN INCORPORATED INTO THE STRUCTURE AS DIRECTED BY THE CONTRACTING OFFICER.
- 4. PER RCW 77.85.050(5E), EACH INDIVIDUAL ROOT WAD AND EACH LOG LARGER THAN TEN FEET LONG AND ONE FOOT IN DIAMETER MUST BE VISIBLY TAGGED WITH A UNIQUE NUMERICAL IDENTIFIER THAT WILL WITHSTAND TYPICAL RIVER CONDITIONS FOR AT LEAST THREE YEARS.

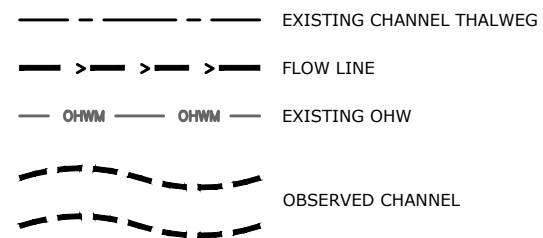


SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP
RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION
GENERAL NOTES
DRAFT PRELIMINARY DESIGN

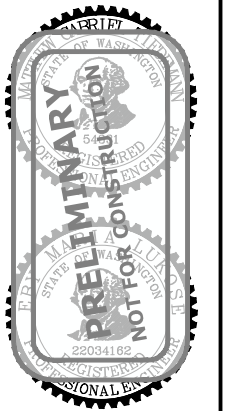
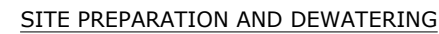
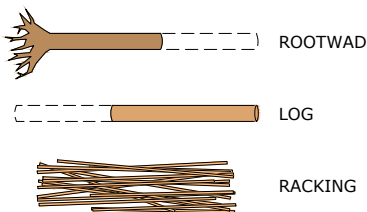
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COUNTY	PIERCE
LATITUDE	46°54'04"N
LONGITUDE	122°00'45"W
TN/SC/RG	T1.6N/S02/R06E
DESIGN MT. EL.	MP
CHECK	SW
CHECK	EL

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IF THIS BAR DOES NOT MEASURE 1" THEN DRAWING IS NOT PLOTTED TO ORIGINAL SCALE.

SITE FEATURES



HABITAT STRUCTURES



LEGEND

DRAFT PRELIMINARY DESIGN

LEGEND

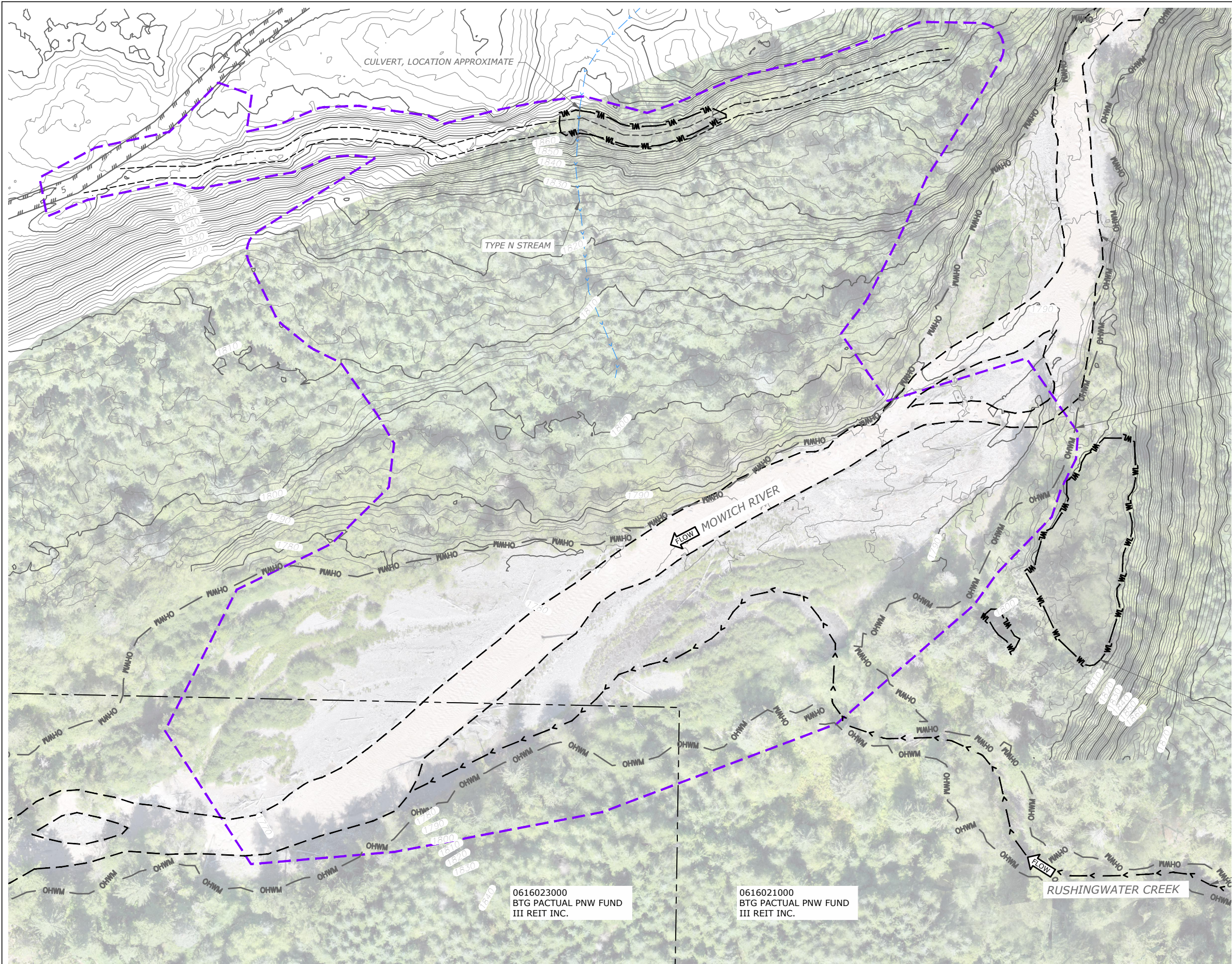
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COUNTY	PIERCE
LATITUDE	46°54'04"N
LONGITUDE	122°00'45"W
TN/SC/RG	T16N/S02/R06E
DESIGN MT, EL	DRAWN MP
CHECK SW	CHECK EL

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IF THIS BAR DOES NOT MEASURE
1" THEN DRAWING IS NOT
PLOTTED TO ORIGINAL SCALE.

SHEET
8 OF 13



- NOTES
- EXISTING CONTOURS ARE SHOWN AS 10' MAJOR, 2' MINOR AND REFLECT 2022 CONDITIONS.
 - AERIAL IMAGERY COLLECTED IN AUGUST 2025 BY SPSSEG.
 - OHWM BASED ON AUGUST 2025 AND MARCH 2026 NSD FIELD OBSERVATIONS, JANUARY 2026 DIGITAL SURFACE MODEL COLLECTED BY SPSSEG, AND 2-YEAR HYDRAULIC MODELING OUTPUT (1,924 CFS IN THE MOWICH RIVER).

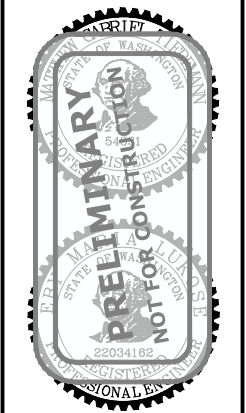
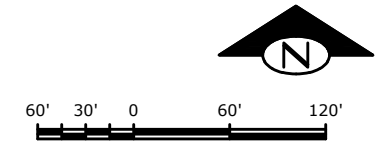
OBSERVED CHANNEL AUGUST 2025

PROJECT LIMITS

DELINEATED WETLAND (TYP)

RUSHINGWATER CREEK FLOWPATH (AUGUST 2025)

EXISTING CONDITIONS
SCALE: 1"=60'



SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP
RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION
EXISTING CONDITIONS PLAN
DRAFT PRELIMINARY DESIGN

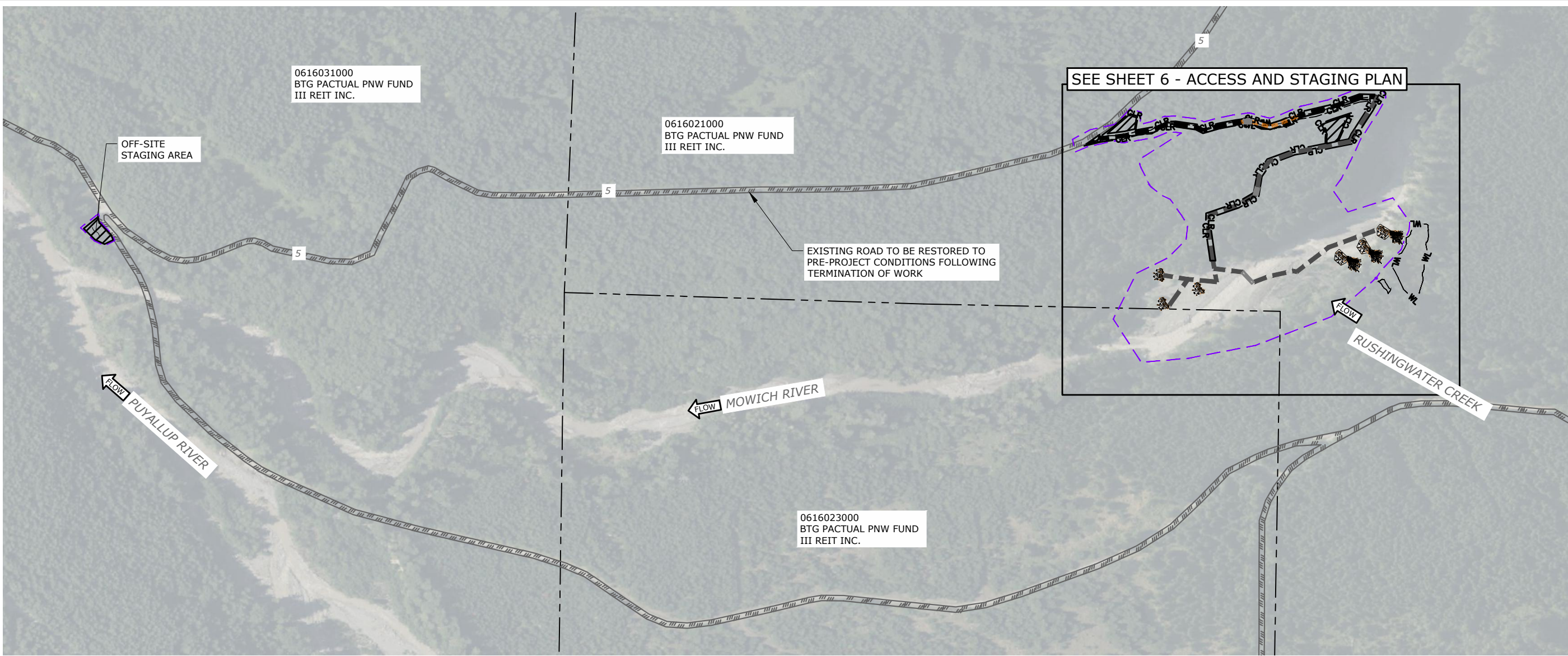
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COUNTY	PIERCE
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LONGITUDE	122°00'45"W
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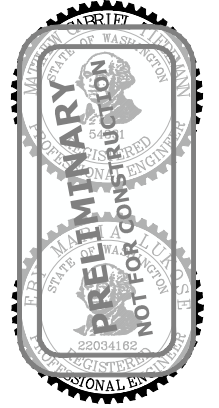
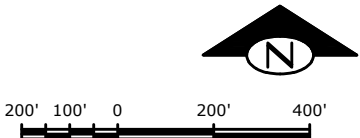
SHEET
4 OF 13

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ACCESS AND STAGING OVERVIEW
SCALE: 1"=200'

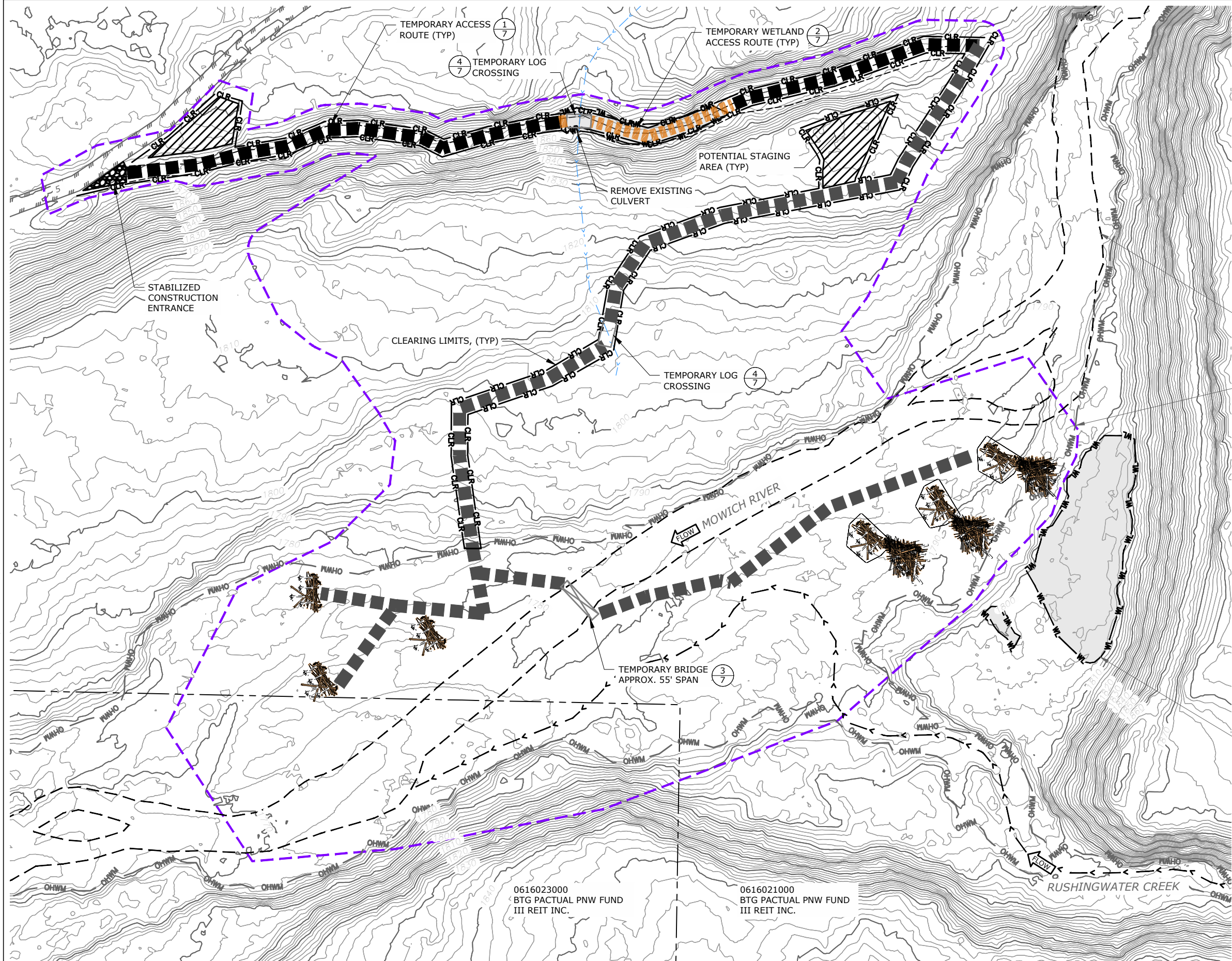


SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP
RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION
ACCESS AND STAGING OVERVIEW
DRAFT PRELIMINARY DESIGN

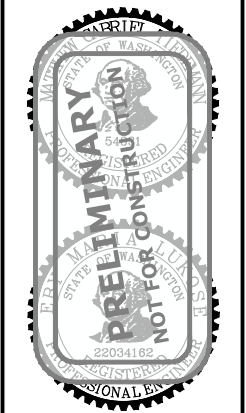
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COUNTY: PIERCE
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LONGITUDE: 122°00'45"W
TN/SC/RG: T16N/S02/R06E
DESIGN MT. EL. DRAWN: MP
CHECK: SW CHECK: EL

IF THIS BAR DOES NOT MEASURE 1" THEN DRAWING IS NOT PLOTTED TO ORIGINAL SCALE.

SHEET
5 OF 13



- NOTES
1. ACCESS ROUTES SHOWN ARE APPROXIMATE AND BASED ON FIELD INVESTIGATIONS CONDUCTED IN AUGUST 2025 AND MARCH 2026. ACCESS ROUTES ARE SUBJECT TO CHANGE BASED ON APPROVAL OF THE ENGINEER OR OWNER.
 2. ACCESS ROUTES TO BE FLAGGED OR STAKED IN THE FIELD AND APPROVED BY THE ENGINEER OR OWNER.
 3. ACCESS ROUTES TO BE RESTORED TO PRE-PROJECT CONDITIONS UPON CONSTRUCTION COMPLETION.



SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP
RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION
ACCESS AND STAGING PLAN
DRAFT PRELIMINARY DESIGN

DATE	APRIL 2026
COUNTY	PIERCE
LATITUDE	46°54'04"N
LONGITUDE	122°00'45"W
TN/SC/RG	TI.6N/S02/R06E
DESIGN MT. EL.	MP
CHECK	SW
CHECK	EL

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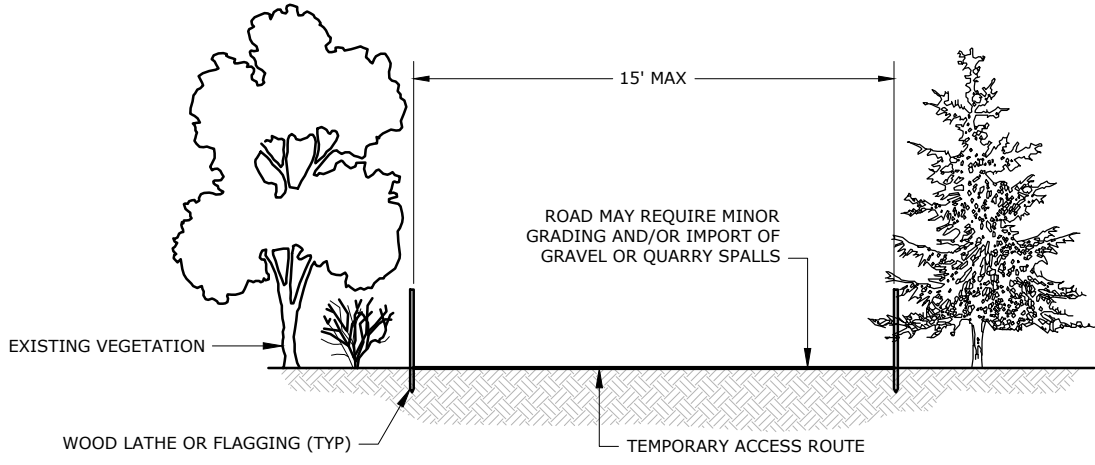
SHEET
6 OF 13

ACCESS AND STAGING PLAN
SCALE: 1"=60'

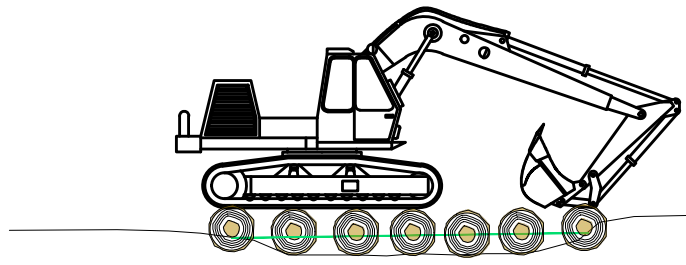


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- NOTES
1. CLEARED ACCESS TO BE ROUTED TO MINIMIZE VEGETATION DISTURBANCE AND FOREST CLEARING.
 2. CONTRACTOR SHALL MARK CLEARING LIMITS WITH FLAGGING. CLEARING LIMITS TO BE APPROVED BY ENGINEER PRIOR TO ANY CLEARING ACTIVITIES.
 3. ANY TREES GREATER THAN 12" Ø SHALL BE REMOVED W/ ROOTWADS INTACT AND STOCKPILED FOR USE IN LOGJAM CONSTRUCTION.
 4. TREES AND SHRUBS WITH 2"-12" Ø SHALL BE STOCKPILED FOR USE AS RACKING MATERIAL IN LOGJAM CONSTRUCTION.
 5. REMAINDER OF VEGETATION AND ORGANIC SOIL SHALL BE GRUBBED, STOCKPILED AND BROADCASTED ON ROAD ALIGNMENT FOLLOWING TERMINATION OF WORK.
 6. ACCESS SHALL BE MAINTAINED BY MINOR GRADING AND IMPORTATION OF WOOD CHIPS, GRAVEL AND/OR QUARRY SPALLS, OR OTHER APPROVED BMPs.
 7. CLEARED ACCESS SHALL BE SCARIFIED AND FULLY DECOMMISSIONED PER DIRECTION OF THE ENGINEER AT THE TERMINATION OF WORK.
 8. REVEGETATION OF DISTURBED AREAS OUTSIDE OF THE APPROVED CLEARING AND ACCESS LIMITS SHALL BE PERFORMED BY CONTRACTOR AT NO COST TO THE OWNER.
 9. ALL GRAVEL OR QUARRY SPALLS PLACED SHALL BE UNDERLAIN WITH A GEOTEXTILE AND REMOVED AT TERMINATION OF WORK IF UTILIZED.



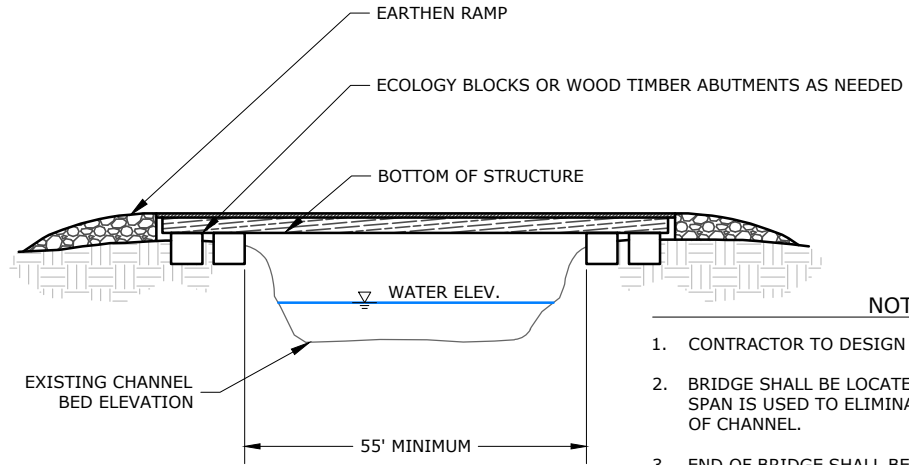
- NOTES
1. CONTRACTOR SHALL PLACE LOGS TO AVOID TRACKING IN WETLAND AND TO MINIMIZE SOIL COMPACTION.
 2. RESTORE WETLAND SURFACE TO PRE-PROJECT CONDITIONS FOLLOWING DECOMMISSIONING OF ACCESS ROAD; REFER TO SITE RESTORATION PLAN ON SHEET 13.

TEMPORARY ACCESS ROUTE
NTS

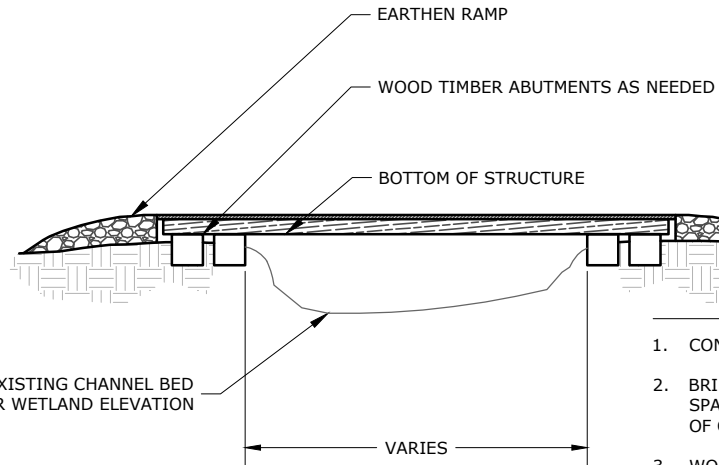
1
7

TEMPORARY WETLAND ACCESS ROUTE
NTS

2
7



- NOTES
1. CONTRACTOR TO DESIGN TEMPORARY BRIDGE.
 2. BRIDGE SHALL BE LOCATED SUCH THAT ONLY ONE SPAN IS USED TO ELIMINATE IMPACTS TO SUBSTRATE OF CHANNEL.
 3. END OF BRIDGE SHALL BEAR ON BANKS WITH SUFFICIENT BEARING CAPACITY TO PREVENT SLOUGHING OR COLLAPSE OF BANKS.
 4. CONCRETE ECOLOGY BLOCKS OR WOOD ABUTMENTS MAY BE USED TO SUPPORT ENDS OF TEMPORARY BRIDGE AS NEEDED.
 5. BRIDGES MAY BE CONSTRUCTED FROM LOGS, RAIL CAR BEDS OR APPROVED EQUAL AND DECKED WITH STEEL SHEET, WOOD LAGGING OR APPROVED EQUAL.
 6. FLOWS IN THE MOWICH RIVER MAY FLUCTUATE DURING EACH DAY DUE TO SNOWMELT. CONTRACTOR TO PROVIDE MINIMUM 1-FT FREEBOARD ABOVE HIGHEST TYPICAL DAILY WATER ELEVATION.



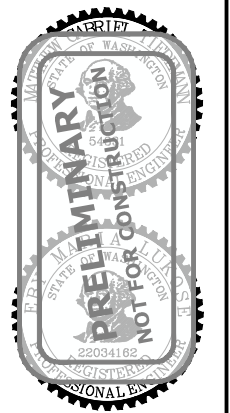
- NOTES
1. CONTRACTOR TO DESIGN TEMPORARY LOG CROSSING.
 2. BRIDGE SHALL BE LOCATED SUCH THAT ONLY ONE SPAN IS USED TO ELIMINATE IMPACTS TO SUBSTRATE OF CHANNEL OR WETLAND.
 3. WOOD ABUTMENTS MAY BE USED TO SUPPORT ENDS OF CROSSING AS NEEDED.
 4. CROSSING MAY BE CONSTRUCTED FROM LOGS, WOOD LAGGING, OR APPROVED EQUAL.
 5. CROSSING SHALL COMPLY WITH WA DNR FOREST PRACTICES PERMIT FOR TEMPORARY CROSSINGS OVER TYPE N STREAMS.

TEMPORARY BRIDGE
NTS

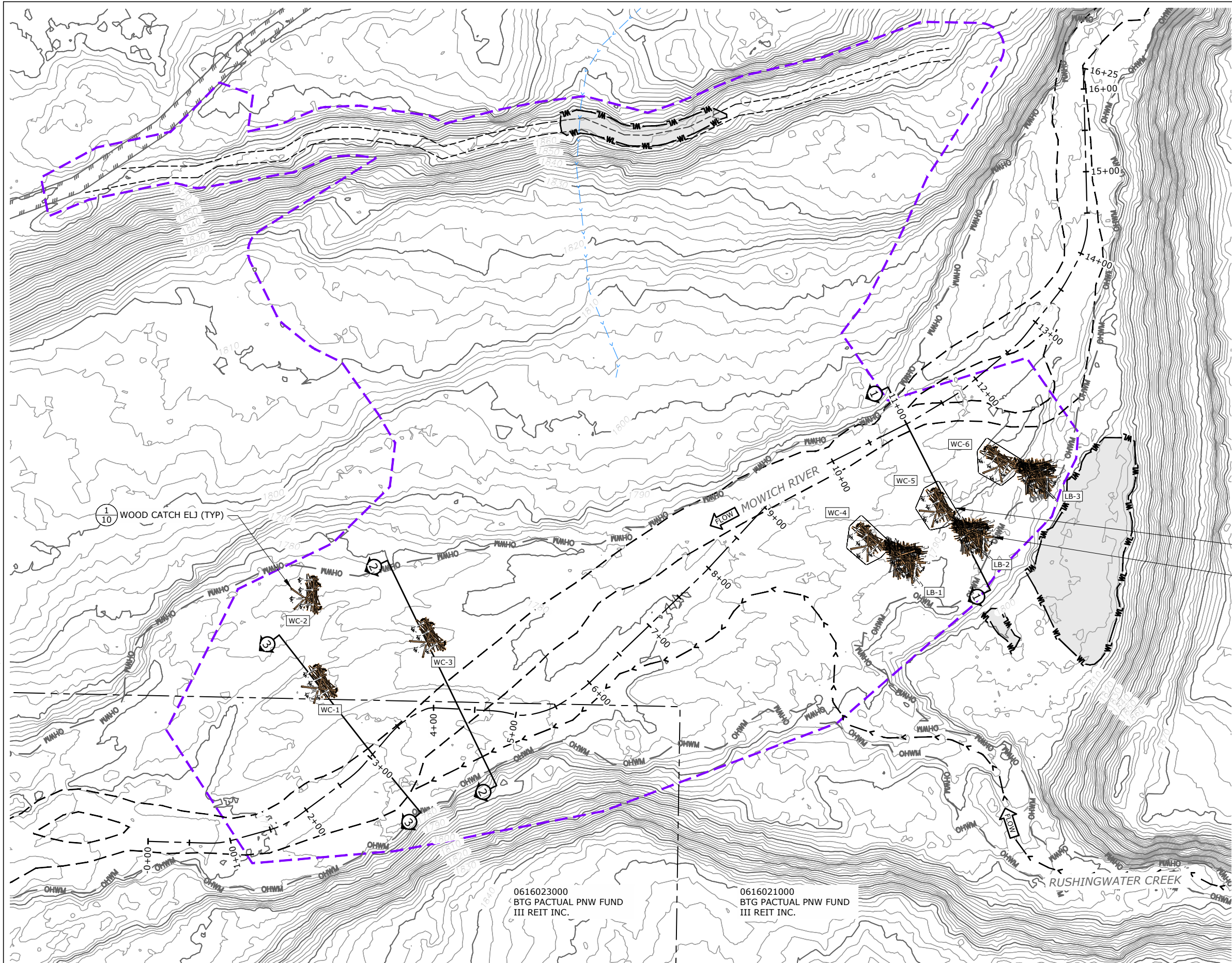
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TEMPORARY LOG CROSSING
NTS

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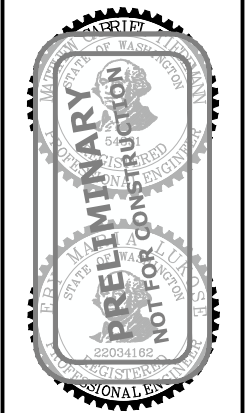
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PROPOSED CONDITIONS
SCALE: 1"=60'

NOTES
1. STRUCTURE LOCATIONS BASED ON 2026 FIELD OBSERVATIONS.

BURIED WOOD CATCH ELJ (TYP) (1/10)
LEFT BANK ELJ (TYP) (1/11)



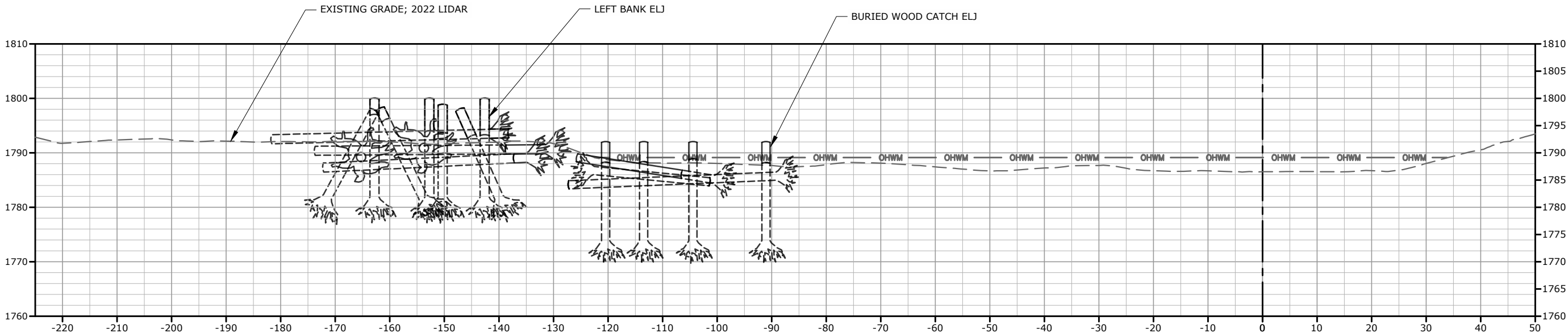
SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP
RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION
PROPOSED CONDITIONS PLAN
DRAFT PRELIMINARY DESIGN

DATE	APRIL 2026
COUNTY	PIERCE
LATITUDE	46°54'04"N
LONGITUDE	122°00'45"W
TN/SC/RG	TI.6N/S02/R06E
DESIGN MT. EL.	DRAWN MP
CHECK SW	CHECK EL

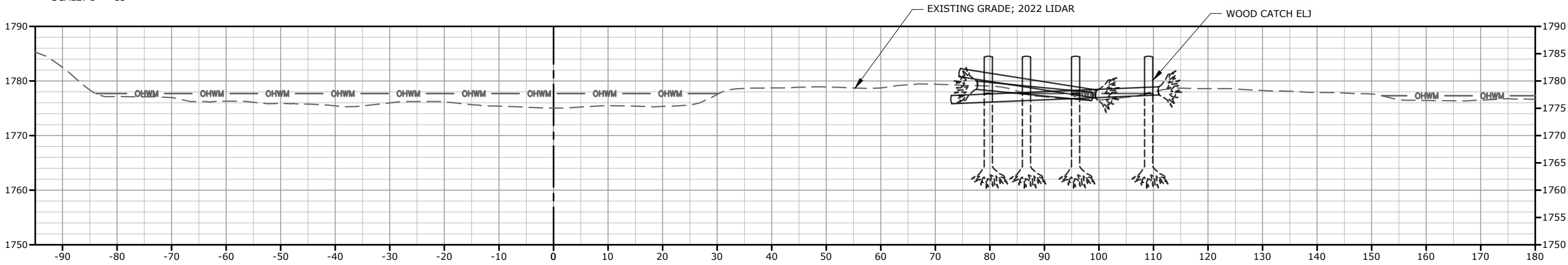
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SHEET
8 OF 13

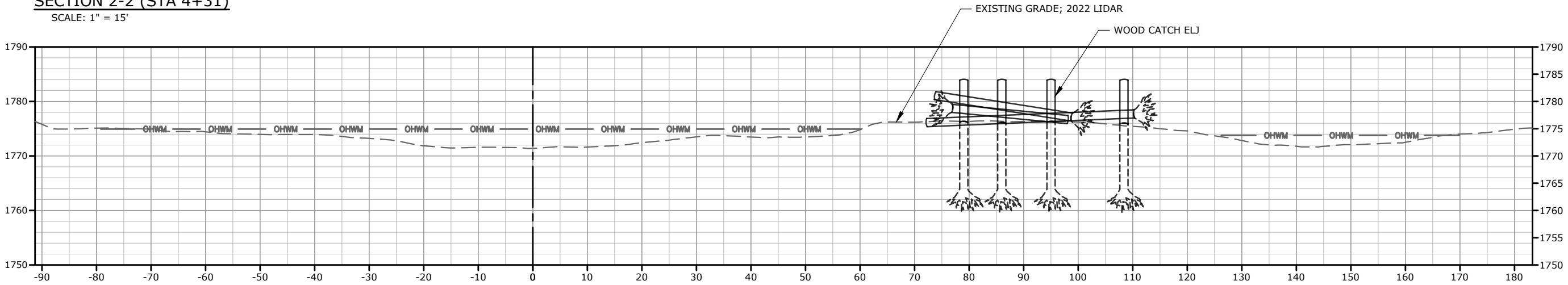
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SECTION 1-1 (STA 11+00)
SCALE: 1" = 15'



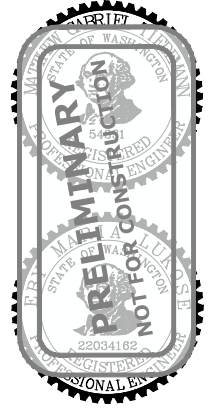
SECTION 2-2 (STA 4+31)
SCALE: 1" = 15'



SECTION 3-3 (STA 3+05)
SCALE: 1" = 15'

NOTES

- ELEVATIONS OF ELJS ARE RELATIVE TO EXISTING GRADE AS REPRESENTED BY 2022 LIDAR. THE MOWICH RIVER IS A DYNAMIC SYSTEM AND HAS CHANGED SINCE THE DATE OF LIDAR COLLECTION. THE CONTRACTOR SHALL ANTICIPATE CHANGES IN THE RIVER PLANFORM BETWEEN BID AND CONSTRUCTION. FINAL ELEVATIONS FOR ALL ELJS SHALL BE BASED ON CONDITIONS AT THE TIME OF CONSTRUCTION AND MAY CHANGE FROM THE ELEVATIONS SHOWN IN THESE PLANS.



SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP

RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION

CROSS SECTIONS

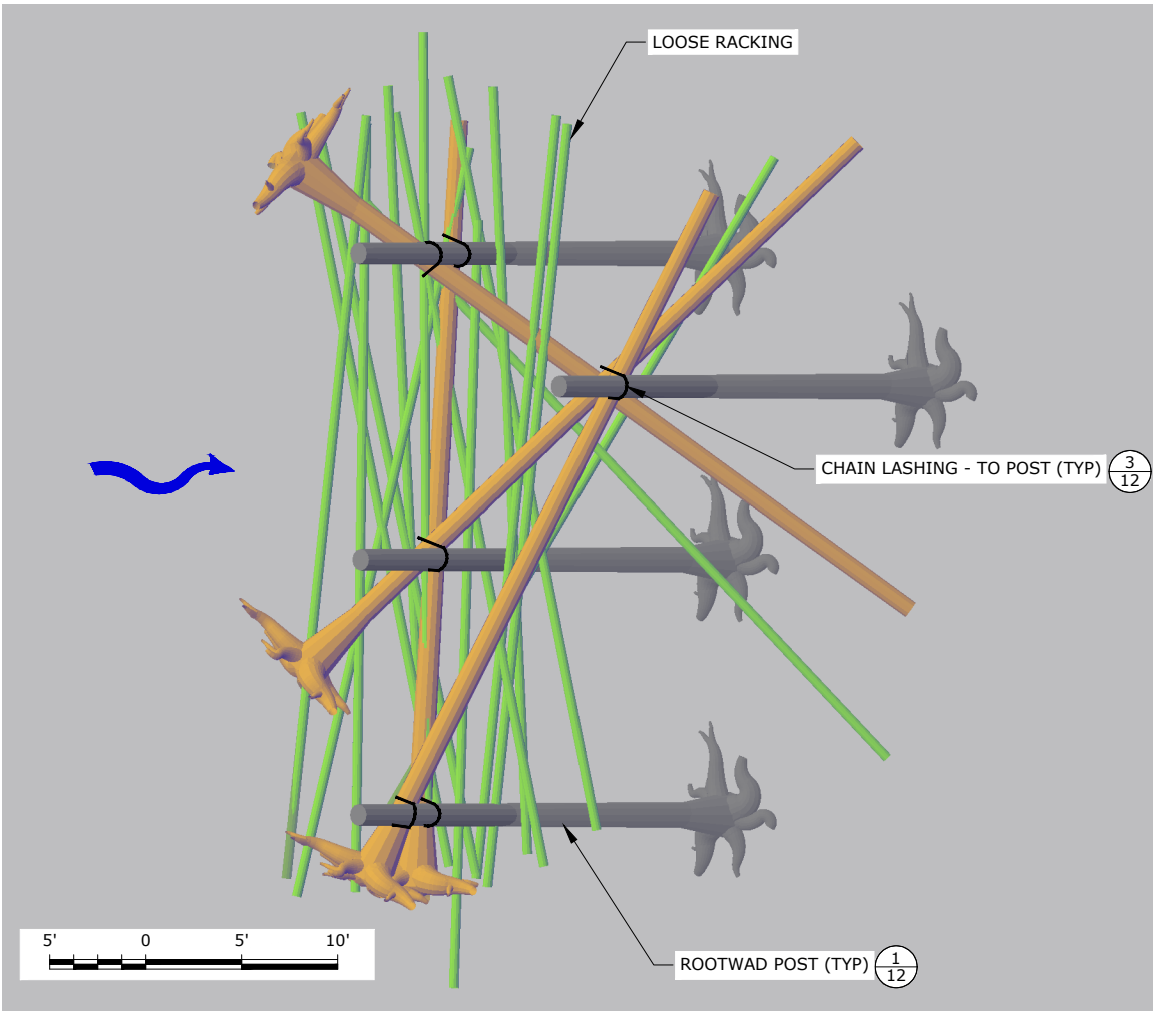
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DESIGN MT. EL.	DRAWN MP
CHECK SW	CHECK EL

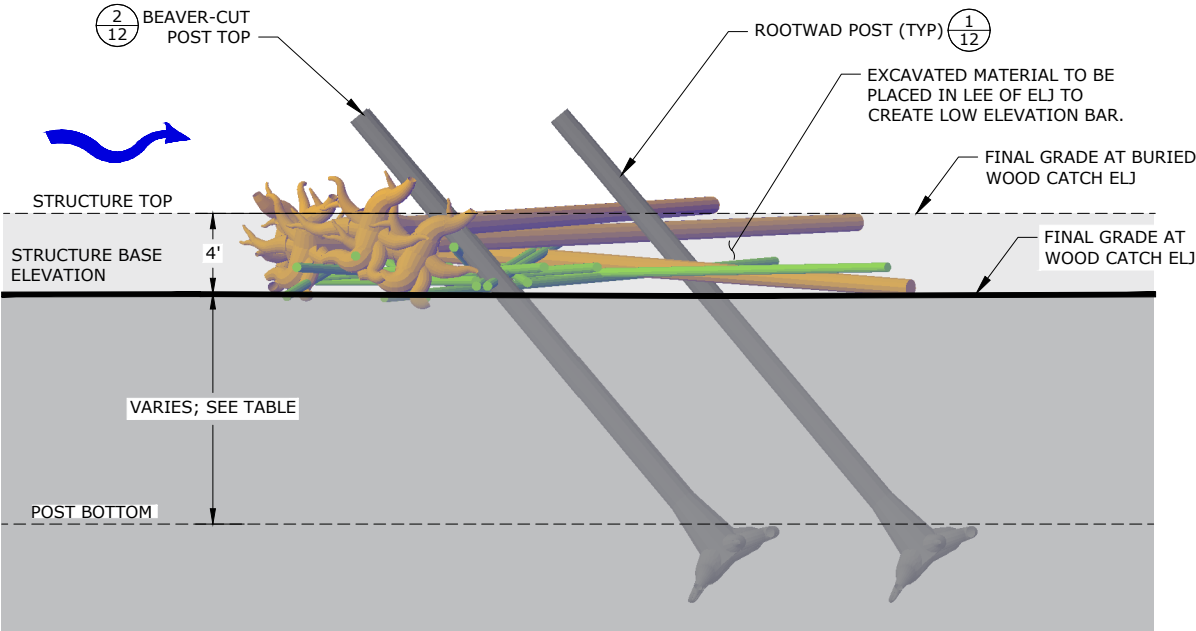
IF THIS BAR DOES NOT MEASURE 1" THEN DRAWING IS NOT PLOTTED TO ORIGINAL SCALE.

SHEET

9 OF 13



WOOD CATCH ELJ PLAN
SCALE: 1" = 5'



WOOD CATCH ELJ PROFILE
SCALE: 1" = 5'

WOOD CATCH ELJ
SCALE: AS SHOWN

NOTES

1. EXCAVATION SPOILS SHALL BE STAGED WITHIN THE WORK AREA AND OUTSIDE FLOWING WATER. SPOILS SHALL BE STOCKPILED TO ALLOW LOG LAYER PLACEMENT AND CONSTRUCTION ACCESS.
2. BACKFILL EXTENTS VARY AND TO BE CONSTRUCTED WITH NATIVE ALLUVIUM FROM EXCAVATION SPOILS.
3. FINAL ELJ HEIGHT TO BE ACHIEVED AS SPECIFIED REGARDLESS OF ACTUAL LOG DIAMETERS USED OR STACKING ARRANGEMENT.
4. ALL LARGE WOOD DIMENSIONS DO NOT INCLUDE BARK THICKNESS.
5. RACKING PLACEMENT SHALL OCCUR WITH EACH LAYER. PLACEMENT TO ENSURE RACKING MATERIAL EXTENDS THROUGH STRUCTURE AND IS PINNED IN PLACE BY SUBSEQUENT LAYERS. A TOTAL OF 30 CY OF SLASH MATERIAL SHALL BE INTERMIXED WITH RACKING PLACEMENTS.
6. THE CONTRACTOR SHALL FIELD VERIFY WITH THE ENGINEER ALL STRUCTURE LOCATIONS, ROOTWAD POST LOCATIONS, LENGTHS, WIDTHS AND ELEVATIONS PRIOR TO EXCAVATION, ASSEMBLY AND INSTALLATION OF EACH STRUCTURE.
7. LOCATIONS FOR ALL STRUCTURE PLACEMENTS WILL BE STAKED IN FIELD BY THE ENGINEER PRIOR TO START OF CONSTRUCTION AT EACH SITE.
8. EXCAVATION LIMITS SHALL BE FIELD VERIFIED BY THE ENGINEER PRIOR TO EXCAVATION COMMENCING AND PLACEMENT OF ANY LARGE WOOD.
9. WOOD PLACEMENT IN EACH LOGJAM LAYER SHALL BE FIELD VERIFIED BY ENGINEER PRIOR TO BACKFILLING.
10. STRUCTURE TOP FOR BURIED WOOD CATCH ELJS SHALL BE AT EXISTING GRADE.

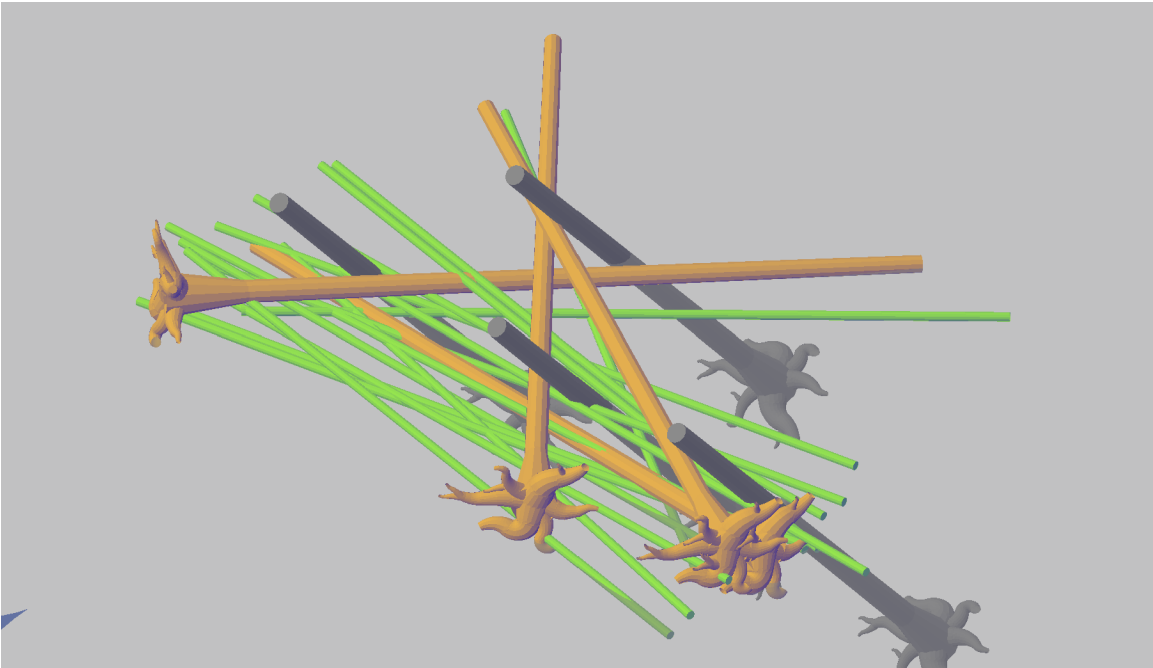
WOOD CATCH ELJ MATERIAL SCHEDULE

ID	MATERIAL	DBH (INCHES)	LENGTH (FEET)	ROOTWAD (Y/N)	QUANTITY PER STRUCTURE
RE-3	ROOTWAD POST	18	30	Y	4
RF-4	ROOTWAD	15	40	Y	4
	LOOSE RACKING	6-12	20-40	OPTIONAL	15
	SLASH	2-6	6-10	N/A	30 CY
	CHAIN LASHING	N/A	VARIES	N/A	6

WOOD CATCH ELJ ELEVATION TABLE

STRUCTURE TYPE	STRUCTURE BASE ELEVATION REFERENCE	POST EMBEDMENT DEPTH (FT)	ANTICIPATED EXCAVATION DEPTH (FT)
WOOD CATCH ELJ	FLOODPLAIN ELEVATION/EXISTING GRADE	13	13
BURIED WOOD CATCH ELJ	4-FEET BELOW EXISTING GRADE	11	15

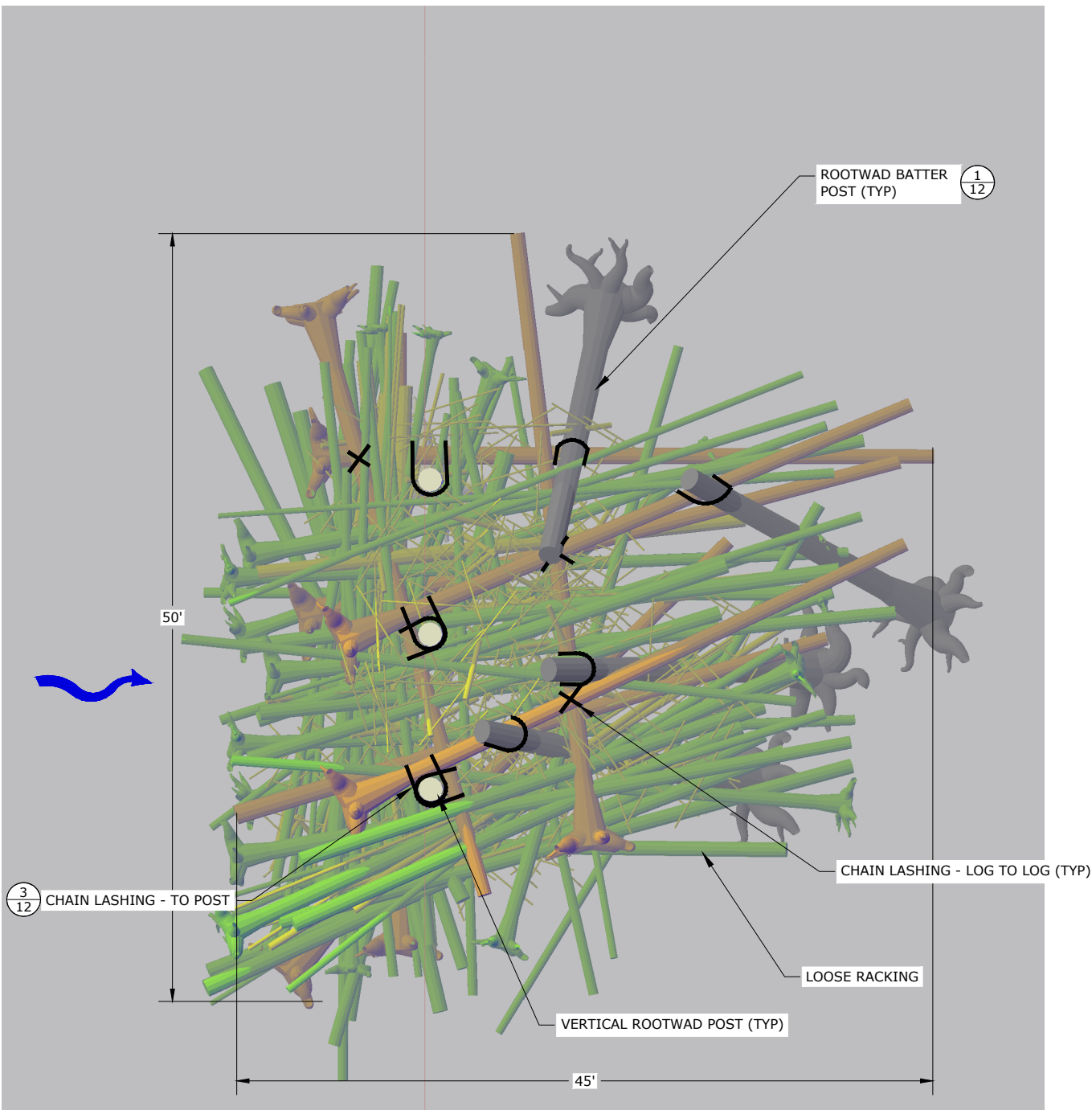
ANTICIPATED EXCAVATION DEPTH IS BASED ON 2022 LIDAR ELEVATIONS.



WOOD CATCH ELJ PERSPECTIVE
NTS

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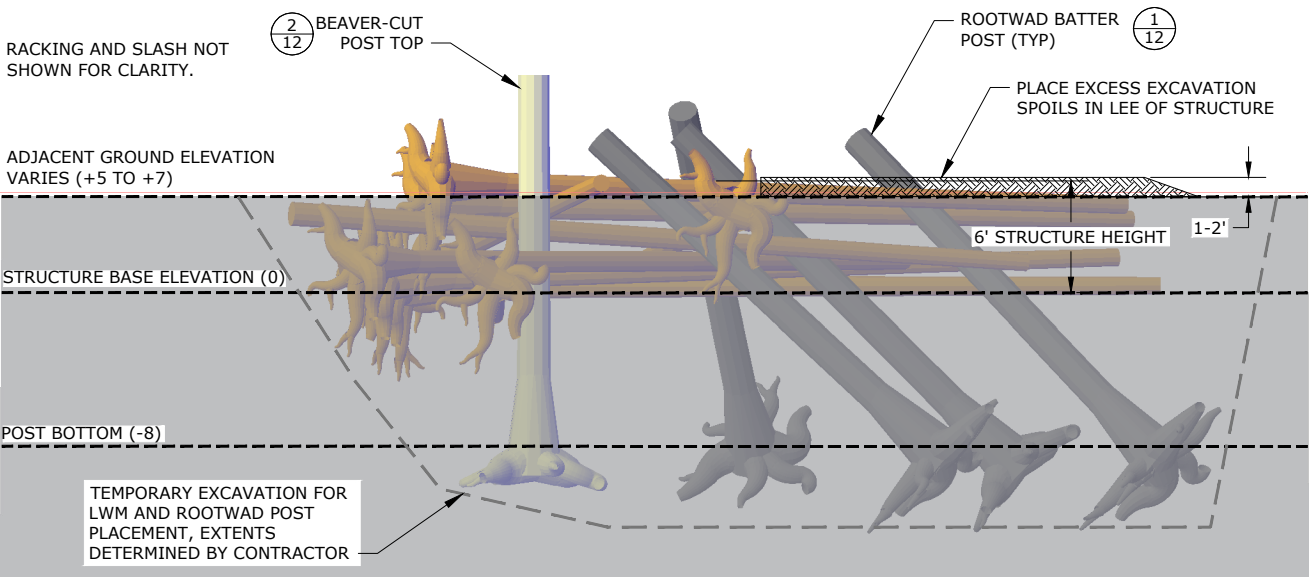
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LEFT BANK ELJ PLAN
SCALE: 1" = 5'



LEFT BANK ELJ
SCALE: AS SHOWN

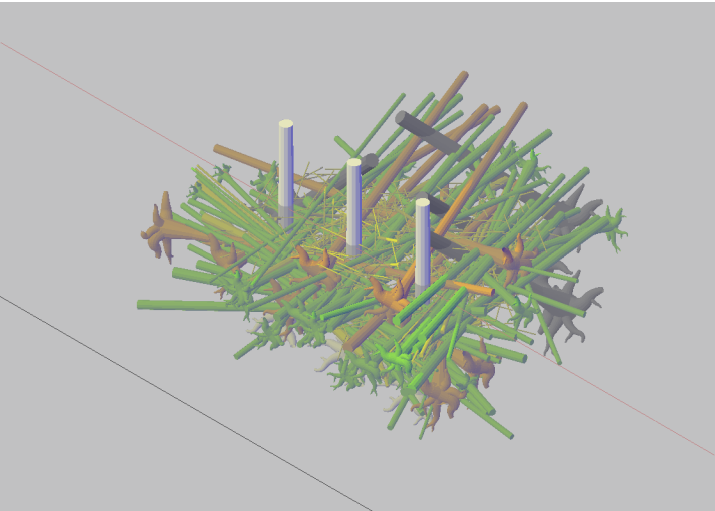


LEFT BANK ELJ PROFILE
SCALE: 1" = 5'

LEFT BANK ELJ MATERIAL SCHEDULE					
ID	MATERIAL	DBH (INCHES)	LENGTH (FEET)	ROOTWAD (Y/N)	QUANTITY PER STRUCTURE
RD-2	ROOTWAD POST	20	25	Y	7
RF-4	ROOTWAD	15	40	Y	8
F-4	LOG	15	40	N	1
	LOOSE RACKING	6-12	20-40	OPTIONAL	150
	SLASH	2-6	6-10	N/A	30 CY
	CHAIN LASHING	N/A	VARIES	N/A	12

LEFT BANK ELJ ELEVATION TABLE			
STRUCTURE TYPE	STRUCTURE BASE ELEVATION REFERENCE	POST EMBEDMENT DEPTH (FT)	ANTICIPATED EXCAVATION DEPTH (FT)
LEFT BANK ELJ	ASSUMED MOWICH THALWEG	8	13-15

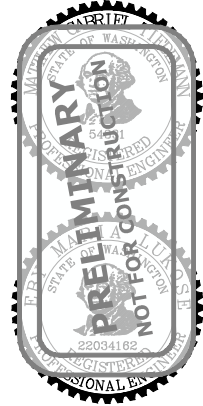
ANTICIPATED EXCAVATION DEPTH IS BASED ON 2022 LIDAR ELEVATIONS.



LEFT BANK ELJ PERSPECTIVE
NTS

NOTES

- EXCAVATION SPOILS SHALL BE STAGED WITHIN THE WORK AREA AND OUTSIDE FLOWING WATER. SPOILS SHALL BE STOCKPILED TO ALLOW LOG LAYER PLACEMENT AND CONSTRUCTION ACCESS.
- BACKFILL EXTENTS VARY AND TO BE CONSTRUCTED WITH NATIVE ALLUVIUM FROM EXCAVATION SPOILS.
- FINAL ELJ HEIGHT TO BE ACHIEVED AS SPECIFIED REGARDLESS OF ACTUAL LOG DIAMETERS USED OR STACKING ARRANGEMENT.
- ALL LARGE WOOD DIMENSIONS DO NOT INCLUDE BARK THICKNESS.
- RACKING PLACEMENT SHALL OCCUR WITH EACH LAYER. PLACEMENT TO ENSURE RACKING MATERIAL EXTENDS THROUGH STRUCTURE AND IS PINNED IN PLACE BY SUBSEQUENT LAYERS. A TOTAL OF 30 CY OF SLASH MATERIAL SHALL BE INTERMIXED WITH RACKING PLACEMENTS.
- THE CONTRACTOR SHALL FIELD VERIFY WITH THE ENGINEER ALL STRUCTURE LOCATIONS, ROOTWAD POST LOCATIONS, LENGTHS, WIDTHS AND ELEVATIONS PRIOR TO EXCAVATION, ASSEMBLY AND INSTALLATION OF EACH STRUCTURE.
- LOCATIONS FOR ALL STRUCTURE PLACEMENTS WILL BE STAKED IN FIELD BY THE ENGINEER PRIOR TO START OF CONSTRUCTION AT EACH SITE.
- EXCAVATION LIMITS SHALL BE FIELD VERIFIED BY THE ENGINEER PRIOR TO EXCAVATION COMMENCING AND PLACEMENT OF ANY LARGE WOOD.
- WOOD PLACEMENT IN EACH LOGJAM LAYER SHALL BE FIELD VERIFIED BY ENGINEER PRIOR TO BACKFILLING.

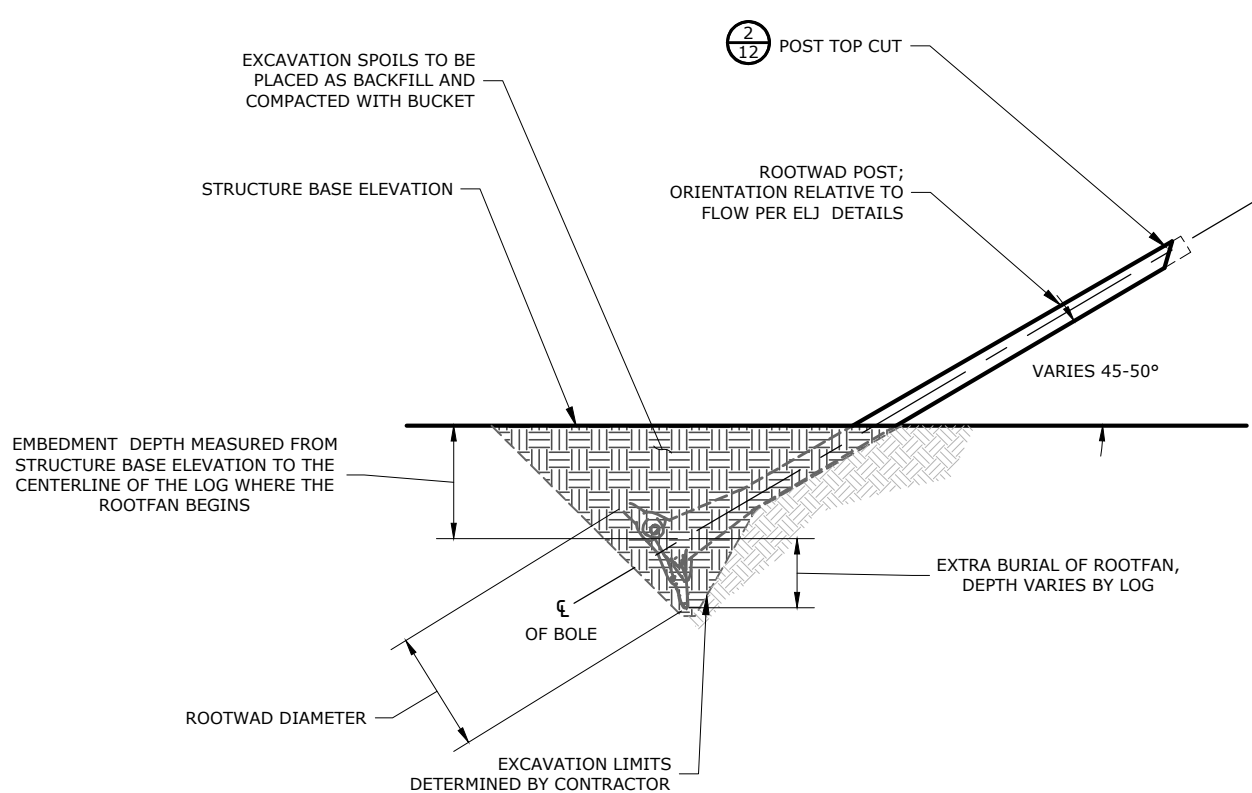


SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP
RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION
LEFT BANK ELJ DETAIL
DRAFT PRELIMINARY DESIGN

DATE	APRIL 2026
COUNTY	PIERCE
LATITUDE	46°54'04"N
LONGITUDE	122°00'45"W
TN/SC/RG	TI 6N/S02/R06E
DESIGN MT. EL.	MP
CHECK	SW
CHECK	EL

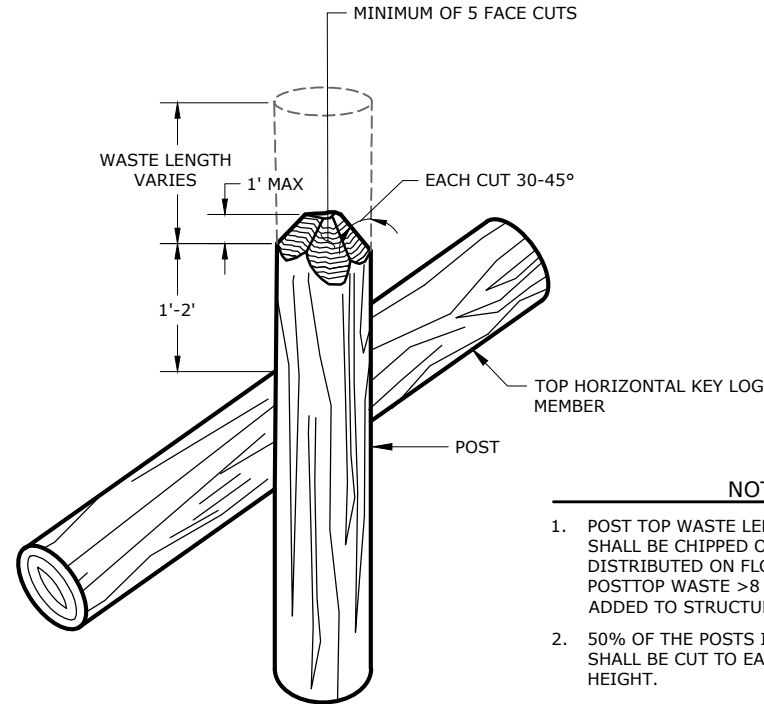
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IF THIS BAR DOES NOT MEASURE 1" THEN DRAWING IS NOT PLOTTED TO ORIGINAL SCALE.

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BATTER POST
SCALE: NTS

1
12

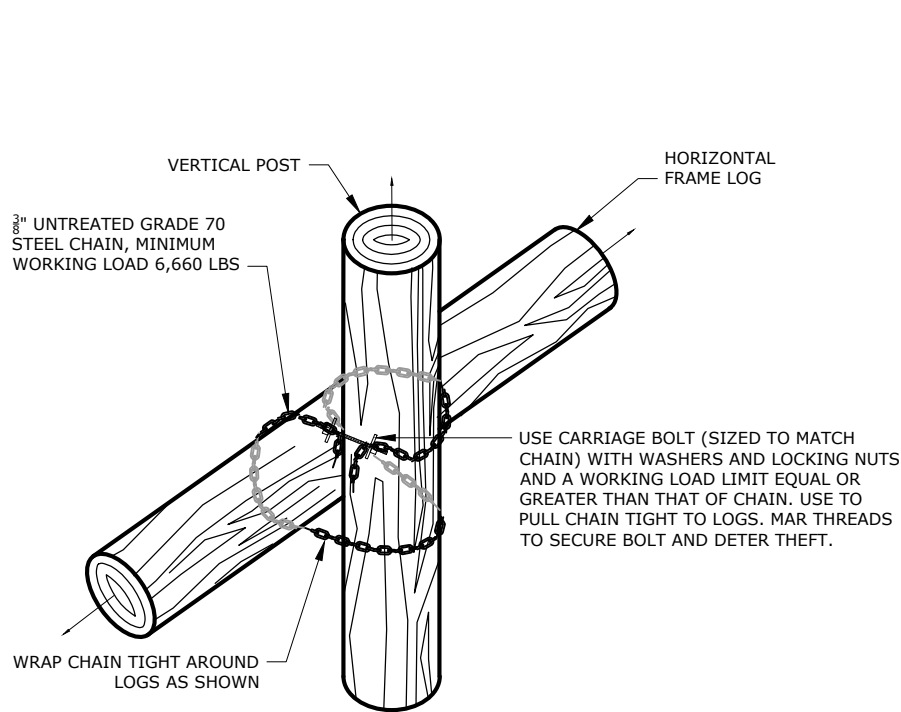


NOTES

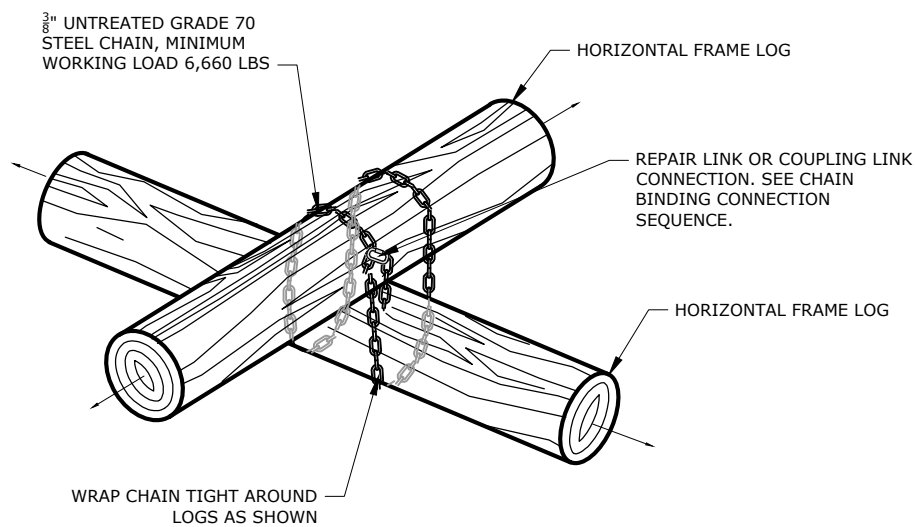
1. POST TOP WASTE LENGTH <8' IN LENGTH SHALL BE CHIPPED OR SHREDDED AND DISTRIBUTED ON FLOODPLAIN AREAS. POSTTOP WASTE >8 FT IN LENGTH CAN BE ADDED TO STRUCTURE AS RACKING.
2. 50% OF THE POSTS IN EACH STRUCTURE SHALL BE CUT TO EACH THE MIN. AND MAX. HEIGHT.

POST TOP
SCALE: NTS

2
12



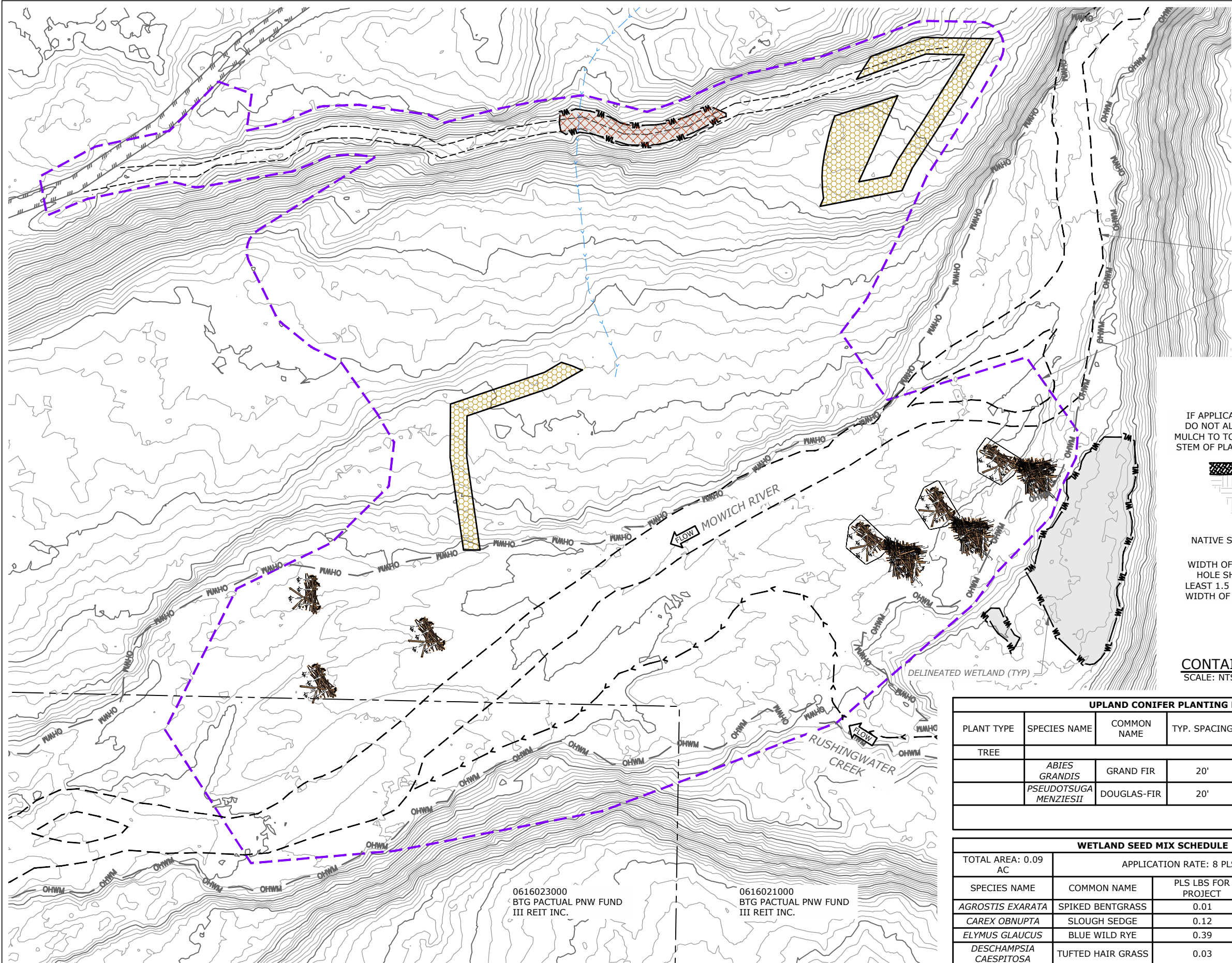
CHAIN LASHING
SCALE: NTS



CHAIN BINDING CONNECTION SEQUENCE

1. USE RATCHET CHAIN BINDER (SIZED TO MATCH CHAIN) TO TIGHTEN CHAIN TO MINIMUM 500 LB TENSION.
2. CONNECT LOOSE ENDS OF CHAIN TOGETHER WITH REPAIR LINK, COUPLING LINK, CARRIAGE BOLT, OR EQUIVALENT.
3. SWITCH RATCHET CHAIN BINDER TO *OUT* SETTING AND SLOWLY RELEASE CHAIN BINDER SO AS NOT TO SHOCK LOAD THE NEWLY CONNECTED CHAIN. REMOVE RATCHET CHAIN BINDER.

3
12



NOTES

1 ACCESS ROUTES TO BE RESTORED TO PRE-PROJECT CONDITIONS UPON CONSTRUCTION COMPLETION.

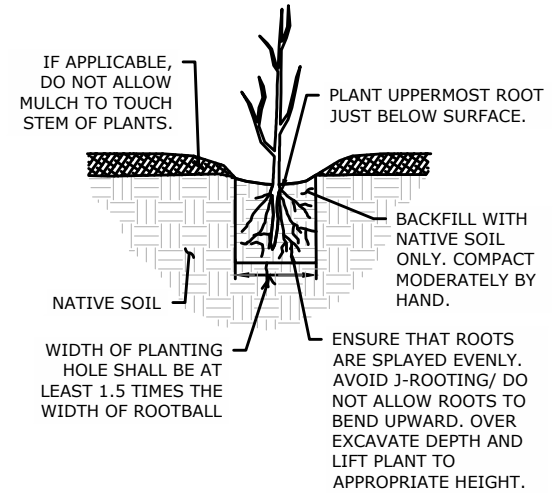
RESTORATION LEGEND

UPLAND CONIFER PLANTING AREA (0.47 ACRES)

WETLAND PLANTING AREA (0.09 ACRES)

OBSERVED CHANNEL AUGUST 2025

PROJECT LIMITS



UPLAND CONIFER PLANTING PLANT SCHEDULE						
PLANT TYPE	SPECIES NAME	COMMON NAME	TYP. SPACING	MATERIAL TYPE	MATERIAL SIZE	TOTAL QUANTITY
TREE	ABIES GRANDIS	GRAND FIR	20'	CONTAINER	#1	10
	PSEUDOTSUGA MENZIESII	DOUGLAS-FIR	20'	CONTAINER	#1	10
	TOTAL QUANTITIES					20

WETLAND SEED MIX SCHEDULE			
TOTAL AREA: 0.09 AC	APPLICATION RATE: 8 PLS LBS/ACRE		
SPECIES NAME	COMMON NAME	PLS LBS FOR PROJECT	% OF MIX
AGROSTIS EXARATA	SPIKED BENTGRASS	0.01	20%
CAREX OBNUPTA	SLOUGH SEDGE	0.12	20%
ELYMUS GLAUCUS	BLUE WILD RYE	0.39	15%
DESCHAMPسيا CAESPITOSA	TUFTED HAIR GRASS	0.03	12%

RESTORATION PLAN
SCALE: 1"=60'



SOUTH PUGET SOUND SALMON ENHANCEMENT GROUP

RUSHINGWATER CREEK ALLUVIAL FAN RESTORATION

RESTORATION PLAN

DRAFT PRELIMINARY DESIGN

DATE: APRIL 2026
COUNTY: PIERCE
LATITUDE: 46°54'04"N
LONGITUDE: 122°00'45"W
TIN/SC/RG: T1.6N/S02/R06E
DESIGN MT. EL. DRAWN: MP
CHECK: SW CHECK: EL

0 1
IF THIS BAR DOES NOT MEASURE 1" THEN DRAWING IS NOT PLOTTED TO ORIGINAL SCALE.

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